SESSION 3 – R^2 SCORE & ADJUSTED R^2

🎯 Topic

- R^2 Score (Coefficient of Determination)
- Adjusted R^2
- Model Evaluation in Regression

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1. 🎯 Learning Objectives

By the end of this session, students will be able to:

- Understand what **R^2 Score** means
- Interpret R^2 values correctly
- Understand limitations of R^2
- Learn why **Adjusted R^2** is needed
- Compare models using R^2 and Adjusted R^2
- Explain answers in **interviews confidently**

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2. 🔁 Quick Recap (Connect Previous Session)

Start class like this:

"In the last session, we learned how to measure model error using MAE, MSE, and RMSE.
But today we will learn another very important metric — R^2 Score, which tells us how well our model explains the data."

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3. 📖 Core Concept: What is R^2 Score?

3.1 Definition

R^2 Score (R-squared) measures how well the regression model explains the variation in the target variable.

3.2 Simple Meaning (Most Important Line)

"R^2 tells us how much of the data our model is able to explain."

3.3 Real-Life Example

Suppose:

We want to predict **Salary based on Experience**

- If model is perfect → it explains salary completely
- If model is poor → it fails to explain variation

Example Interpretation:

R^2 Value	Meaning
1.0	Perfect model

R^2 Value	Meaning
0.8	80% data explained
0.5	Only 50% explained
0	Model is useless
Negative	Worse than average

3.4 Intuition (Very Important)

Imagine:

- Data points scattered
- Model tries to fit a line

👉 R^2 measures:

"How close are predictions to actual values compared to a simple average model?"

4. 📊 Concept Behind R^2 (Simple Intuition)

We compare 2 things:

1. Total Variation (TSS)

Variation in actual data

2. Residual Variation (RSS)

Error made by model

Formula Idea (No heavy maths)

$$R^2 = 1 - \frac{Error}{TotalVariation}$$

Meaning:

- If error is small → R^2 close to 1
- If error is large → R^2 close to 0

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5. 📌 Important Properties of R^2

✅ **Property 1**

Range:

$$(-\infty, 1]$$

✅ **Property 2**

Higher R^2 = Better model (generally)

✅ **Property 3**

R^2 can be **negative**

👉 Very important for interview

Example:

If model performs worse than mean prediction → R^2 becomes negative

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6. ❗ Limitations of R^2

This is **VERY IMPORTANT**

Problem 1: R^2 always increases with more features

Even if feature is useless:

- R^2 may increase

👉 This is misleading

Problem 2: Does not detect overfitting

Model may:

- Fit training data well
- Perform badly on new data

But R^2 still high

Teaching Line

"R^2 alone cannot be trusted for model selection."

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7. 📌 Solution: Adjusted R^2

7.1 Definition

Adjusted R^2 improves R^2 by penalizing unnecessary features.

7.2 Simple Meaning

"Adjusted R^2 only increases if the new feature actually improves the model."

7.3 Why Needed?

Because:

- R^2 increases blindly
- Adjusted R^2 is smarter

7.4 Formula (Just Show, Don't Derive)

$$Adjusted\ R^2 = 1 - \left(\frac{(1 - R^2)(n - 1)}{n - k - 1}\right)$$

Where:

- n = number of data points
- k = number of features

7.5 Key Difference

Metric	Behavior
R^2	Always increases
Adjusted R^2	Increases only if feature is useful

8. 📊 Real-Life Example (Important)

Suppose:
You add a useless feature like:

- Random number column

Result:

- R^2 increases ❌
- Adjusted R^2 decreases ✔

Teaching Line

"Adjusted R^2 protects us from adding useless features."

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9. 📊 When to Use R^2 vs Adjusted R^2

Use R^2 when:

- Single feature model
- Quick evaluation

Use Adjusted R^2 when:

- Multiple features
- Feature selection
- Model comparison

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10. 🎯 Interview Concepts (Very Important)

Q1:

What is R^2 Score?

👉 It measures how well the model explains variance in data.

Q2:

Can R^2 be negative?

👉 Yes, when model performs worse than baseline.

Q3:

Why Adjusted R^2 is better?

👉 Because it penalizes unnecessary features.

Q4:

Difference between R^2 and Adjusted R^2?

👉 R^2 always increases, Adjusted R^2 increases only when feature is useful.

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11. 💻 PRACTICAL (Your Format)

🎯 **Objective**

Compare:

- Linear Regression
- Polynomial Regression

Using:

- R^2
- Adjusted R^2

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12. 📊 Output Interpretation

Tell students:

Observation:

- Polynomial Regression has higher R^2
- Adjusted R^2 also higher

👉 Conclusion:

Polynomial model explains data better

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13. ⚠️ Important Teaching Point

Ask students:

"If we increase polynomial degree to 5, will R^2 increase?"

Answer:
👉 Yes

Then ask:

"Should we always do that?"

Answer:
👉 No → overfitting

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14. 🧪 Practice Task

- Change degree to 3
- Calculate R^2 and Adjusted R^2
- Compare results

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15. 🏠 Homework

- Write difference between R^2 and Adjusted R^2
- Why R^2 alone is misleading?
- Try model with extra useless feature

"R^2 tells us how well our model explains the data, but it can be misleading when we add more features.
Adjusted R^2 helps us make better decisions by penalizing unnecessary complexity.
A good model is not just accurate, but also simple and meaningful."